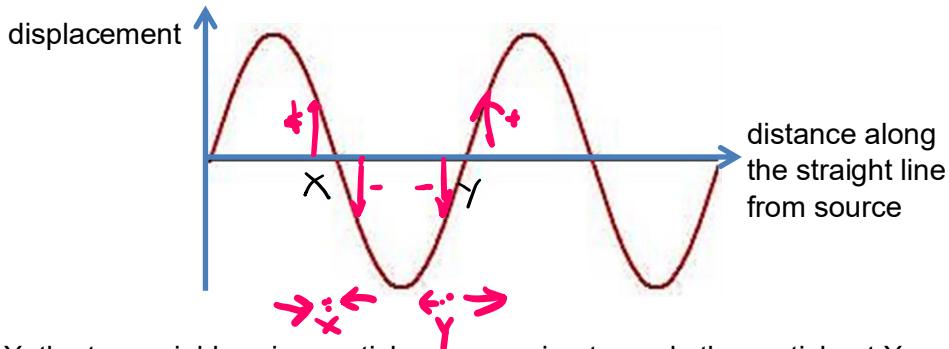


$$\text{Max } v = x_0 \omega = 0.030 \times (2\pi/2.0) = 0.094 \text{ m/s}$$

16 B



At X, the two neighbouring particles are moving towards the particle at X, so it is a compression region. At Y the two neighbouring particles are moving away from the particle at Y.

17 A

Rayleigh's criterion: minimum angle of resolution $\theta = \lambda/b$